

Module -1

DC Circuit Analysis and Theorems

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Mesh Analysis

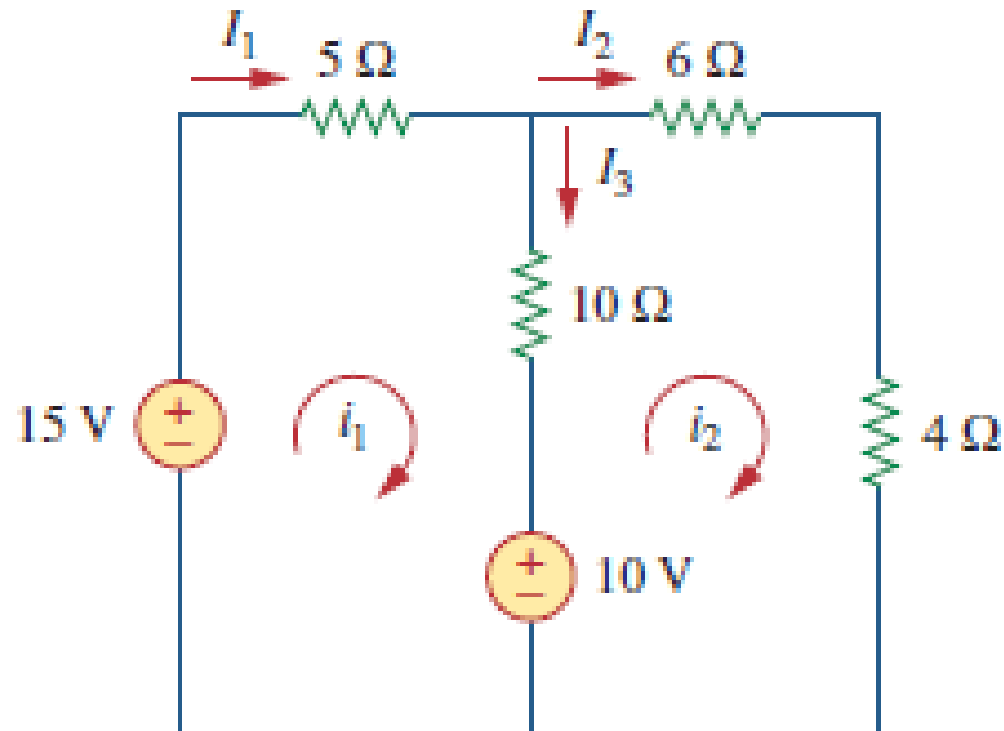
A mesh is a loop which does not contain any other loops within it.

Steps:

1. Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

Tutorial - 1

Find the branch currents I_1 , I_2 and I_3 using mesh analysis.



Solution:

We first obtain the mesh currents using KVL. For mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

$$3i_1 - 2i_2 = 1$$

For mesh 2,

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

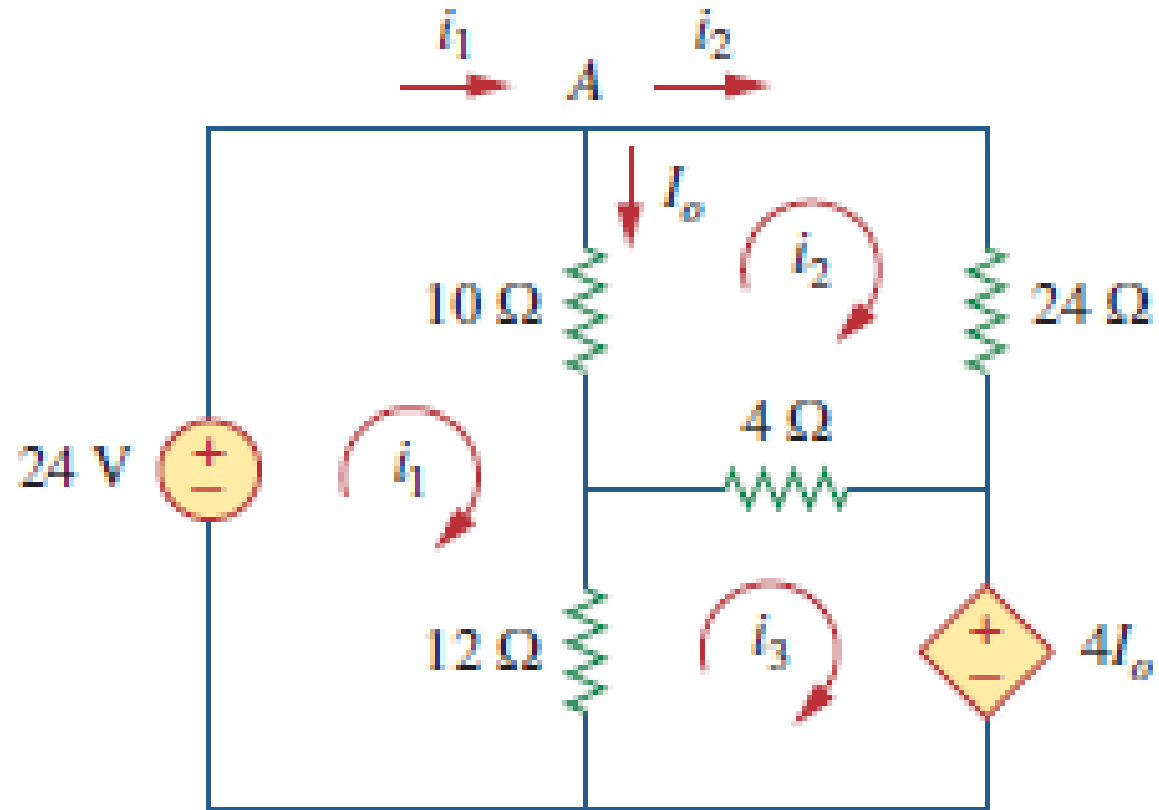
$$i_1 = 2i_2 - 1$$

$$6i_2 - 3 - 2i_2 = 1 \quad \Rightarrow \quad i_2 = 1 \text{ A} \quad i_1 = 2i_2 - 1 = 2 - 1 = 1 \text{ A.}$$

$$I_1 = i_1 = 1 \text{ A}, \quad I_2 = i_2 = 1 \text{ A}, \quad I_3 = i_1 - i_2 = 0$$

Tutorial - 2

Determine I_o in the circuit using mesh analysis.



Solution:

We apply KVL to the three meshes in turn. For mesh 1

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

or

$$11i_1 - 5i_2 - 6i_3 = 12$$

For mesh 2,

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$$

or

$$-5i_1 + 19i_2 - 2i_3 = 0$$

For mesh 3,

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

But at node A, $I_o = i_1 - i_2$, so that

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

or

$$-i_1 - i_2 + 2i_3 = 0$$

We calculate the mesh currents using Cramer's rule as

$$i_1 = \frac{\Delta_1}{\Delta} = \frac{432}{192} = 2.25 \text{ A}, \quad i_2 = \frac{\Delta_2}{\Delta} = \frac{144}{192} = 0.75 \text{ A},$$

$$i_3 = \frac{\Delta_3}{\Delta} = \frac{288}{192} = 1.5 \text{ A}$$

Thus, $I_o = i_1 - i_2 = 1.5 \text{ A}$.

Supermesh Analysis

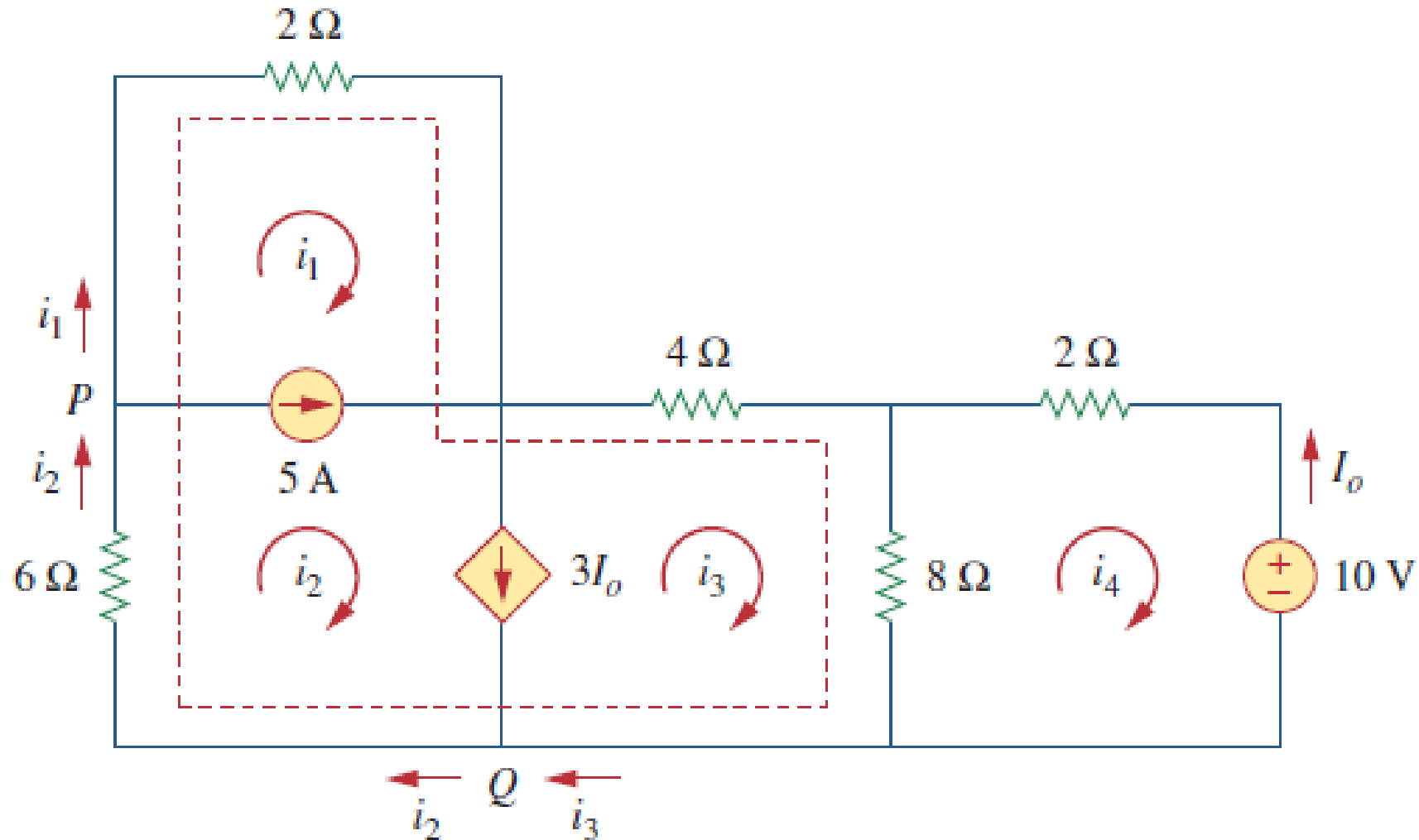
A supermesh results when two meshes have a (dependent or independent) current source in common.

Case 1: When a current source exists only in one mesh, we can write the mesh current as the source current.

Case 2: When a current source exists between two meshes, that is referred as a Supermesh. Apply both KCL and KVL to determine the branch currents.

Tutorial - 3

- Find i_1 to i_4 using mesh analysis.



Solution:

Applying KVL to the larger supermesh,

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

$$i_1 + 3i_2 + 6i_3 - 4i_4 = 0$$

For the independent current source, we apply KCL to node P :

$$i_2 = i_1 + 5$$

For the dependent current source, we apply KCL to node Q :

$$i_2 = i_3 + 3I_o$$

But $I_o = -i_4$, hence,

$$i_2 = i_3 - 3i_4$$

Applying KVL in mesh 4,

$$2i_4 + 8(i_4 - i_3) + 10 = 0$$

$$5i_4 - 4i_3 = -5$$

$$i_1 = -7.5 \text{ A}, \quad i_2 = -2.5 \text{ A}, \quad i_3 = 3.93 \text{ A}, \quad i_4 = 2.143 \text{ A}$$